

The Agenda Thus Far

1. $f: 2^E \rightarrow \mathbb{R}$ monotone submodular

2. $\max: f(I)$
st: $I \in \mathcal{I}$, $\mathcal{I}$ matroid

3. $F(x) = \sum_{S \subseteq E} f(S) \cdot \prod_{i \in S} x_i \cdot \prod_{j \notin S} (1-x_j) = \mathbb{E}_x[f(S)]$

$\forall x \in \{0,1\}^E$, $F(x) = f(x)$

Hence: $\text{OPT} = \max_{\text{st: } x \in \mathcal{P}(\mathcal{I})} F(x) \geq \max_{\text{st: } I \in \mathcal{I}} f(I)$

4. for $x(t)$ the output of the Continuous Greedy alg:

$f(I^*) = F(I^*) \geq F(x(t)) \geq (1 - 1/e) \cdot \text{OPT} \geq (1 - 1/e) \cdot f(I^*)$
 $\uparrow$
 $F(x^*)$

This Lecture: Main Result

Thm: $\exists$ efficient rounding procedure
s.t. $\forall$ subm. fn $f: 2^E \rightarrow \mathbb{R}$, matrix $\mathcal{X}$
at any $x_0 \in P(\mathcal{M})$, outputs $S \in \mathcal{Z}$
s.t. $f(S) \geq F(x_0)$.

Note: if f, F linear this result
would be immediate:

$$x_0 \in P(\mathcal{M}) \Rightarrow x_0 = \sum \alpha_i x^i$$

$$F(x) = \sum \alpha_i F(x^i) = \sum \alpha_i f(x)$$

Proof — 1/4

We assume: $x_0 \in P_{\text{base}}(K)$

Since $x_0 = x(1)$ is the output of (the discretization of)
the cont. greedy alg, we have from the alg.
itself: $x_0 = \frac{1}{N} \sum_{i=1}^N x^{B_i}$

We will combine bases $B_1, \dots, B_N$
involved in the decomposition of x_0 ,
and use convexity properties of F .

For simplification of notation: $N = 2^q$

Proof — 2/4

key sub-lemma: Merge Bases

Input: B_1, B_2

While: $B_1 \neq B_2$

Find $i \in B_1 \setminus B_2, j \in B_2 \setminus B_1$ s.t.

$$B_1 - i + j \in \mathcal{L}, B_2 + i - j \in \mathcal{L}$$

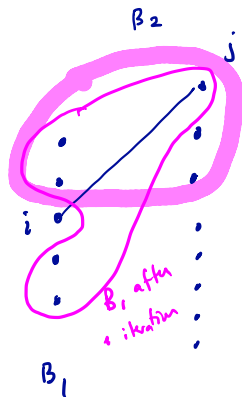
$$\begin{aligned} \text{w. prob } \frac{1}{2}: \quad B_1 &\leftarrow B_1 - i + j \\ \text{or} \quad B_2 &\leftarrow B_2 + i - j \end{aligned}$$

Output $B_1 = B_2 = B_1 \pm (i_1 - j_1) \pm (i_2 - j_2) \pm \dots$

Note: i, j participate only once

$i \in B_1 \cup B_2$
↑
updated

$j \in B_1 \cap B_2$
↑
updated



Proof — 3/4

Lemma: $x_0 = \frac{1}{N} \sum x^{B_i}$, pick B_1, B_2
 $B = \text{Merge Basins } (B_1, B_2)$
 $\uparrow$
 random

$$\mathbb{E} \left[F \left(x - \frac{1}{N} (x^{B_1} + x^{B_2}) + \frac{2}{N} x^B \right) \right] \geq F(x_0)$$

pf: $\rightarrow \mathbb{E}_{\xi_1, \dots, \xi_\ell} \left[F \left(x + \frac{1}{N} \xi_1 (e_i - e_j) + \dots + \frac{1}{N} \xi_\ell (e_i - e_j) \right) \right]$

$$= \mathbb{E}_{\xi_1, \dots, \xi_{\ell-1}} \left[\underbrace{\mathbb{E}_{\xi_\ell} \left[F \left(x + \frac{1}{N} \sum_{k=1}^{\ell-1} \xi_k (e_i - e_j) + \frac{1}{N} \xi_\ell (e_i - e_j) \right) \right]}_{\text{Recall: } F \text{ is convex along } d = e_i - e_j} \middle| \xi_1, \dots, \xi_{\ell-1} \right]$$

$$\geq \mathbb{E} \left[F \left(x + \frac{1}{N} \sum \xi_k (e_i - e_j) \right) \middle| \dots \right]$$

$$\geq F(x)$$

$$\rightarrow \mathbb{E} \left[F \left(x + \frac{1}{N} \xi (e_i - e_j) \right) \right] = \frac{1}{2} F \left(x + \frac{1}{N} (e_i - e_j) \right) + \frac{1}{2} F \left(x - \frac{1}{N} (e_i - e_j) \right) \geq F(x),$$

Proof — 4/4

$$N = 2^q.$$

Pick pairs and continue merging distinct pairs until we have $N/2$ boxes.

Repeat until we have a single box B^* .

This satisfies:

$$\mathbb{E}[f(B^*)] \geq F(x_0)$$